# Preseason Screen Cannot Predict Injury over Three Years of College Football

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**ABSTRACT**

MORTENSEN, B. B., U. H. MITCHELL, A. W. JOHNSON, G. W. FELLINGHAM, J. B. FELAND, and J. W. MYRER. Preseason Screen Cannot Predict Injury over Three Years of College Football. *Med. Sci. Sports Exerc.*, Vol. 52, No. 11, pp. 2286–2292, 2020. **Purpose:** This study aimed to investigate if the Functional Movement Screen (FMS) total score, individual component test scores, or number of asymmetries can predict noncontact injury risk over three consecutive seasons of National Collegiate Athletic Association Division I football. **Methods:** Because football teams comprise individuals with vastly different physical characteristics and playing responsibilities, we divided the subjects into three homogeneous groups based on position (big, combo, and skill). Each FMS score was assessed with regard to the total team score and by individual position groups. For our injury analysis, we also controlled for exposure. Two hundred and eight National Collegiate Athletic Association Division I athletes participated over three consecutive seasons, yielding a total of 343 observations. **Results:** There was no significant relationship between total FMS score and likelihood of injury when analyzed by the total team or by position group. These findings were the same for all groups, for both the total number of injuries and injuries weighted by injury exposure. The only significant findings occurred when we considered individual test item scores to injury by position group. We only found a significant relationship in the expected direction with push-up stability in the combo group. **Conclusions:** FMS was not a good predictor of noncontact injury. **Key Words:** FUNCTIONAL MOVEMENT SCREEN, FMS, INJURY PREDICTION, COLLEGE FOOTBALL

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Participation in athletics has always carried an inherent risk for injury. American football is not only noted to be the most popular high school and collegiate sport, with a total overall participation of 9 million athletes, but also the leading cause of sports-related injuries (1). Intercollegiate football has been shown to have the highest injury rate of National Collegiate Athletic Association (NCAA) sports, with a reported 35.9 injuries per 1000 injury exposures (IE) (2,3). Likewise, studies estimate that high school football players suffer between 300,000 and 1.2 million football-related injuries annually (1,3). The mean cost for care of approximately 400,000 visits to the emergency room due to football injury was estimated to be $2000 per injury, whereas the total cost of football-related emergency room visits in the United States over a 2-yr period was over $770,000,000 (1,3). With the goal to improve athletes' safety, the NCAA and its member institutions have put considerable efforts into studying and consequently changing practice policies and game rules and mandating improved equipment standards (2,4–9).

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Several factors associated with an increased risk of injury in football players have been identified, for example, large body mass (10), previous injury (11), and overuse (12). To date, there is no one standardized screening tool that successfully identifies additional predisposing risk factors. Currently, football teams evaluate a wide variety of characteristics in which the athletes have to excel, such as muscle power, speed, flexibility, and endurance (13). In addition, over the past 10 yr, it has become common for many teams to also utilize the Functional Movement Screen (FMS) as a preparticipation screening tool in order to determine whether the athlete has the essential movements needed to participate in sports activities at a level of minimum competency (14). The FMS is an inexpensive, quick, and easy to administer set of seven tests that are designed to identify restrictions or alterations to normal basic movement patterns (14,15). The seven FMS movements are as follows: deep squat, hurdle step, inline lunge, shoulder mobility, active straight leg raise, trunk stability push-up, and rotary stability in quadruped position. Scoring ranges from 0 to 3. A score of 0 indicates that pain was associated with the movement. Otherwise, a score of 1 to 3 is given based on the quality of movement. A maximum score on the seven test items is 21. Information on asymmetry in either range of motion or strength is obtained by the five FMS tests that compare right-to-left differences, that is, hurdle step, inline lunge, shoulder mobility, active straight leg raise, and rotary stability in quadruped position.

Over the past 11 yr, numerous studies have attempted to establish the efficacy of the FMS for injury prediction. Results to



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this point have been mixed. Some studies have reported an association in an athlete's FMS score and the athletes' chance of injury (16–18). Conversely, other studies have reported no relationship between scores and injury prediction (19,20), and others have challenged the utility of a total FMS score, reporting a lack of internal consistency in FMS sum scores (21). In addition, research discussing FMS in regard to football injuries is based on a small sample of professional players from one or two teams over a season or preseason (16,17). As only 1.5% of all NCAA football players go on to participate in the NFL, the results of those studies may not be generalizable to the typical NCAA level football athlete (22).

Because the FMS was introduced as a way to systematically assess basic movement pattern quality with regard to movements, which are less than optimal or problematic (14), we decided to concentrate on the correlation of the FMS with noncontact injury. We did this on the assumption that movement quality likely has considerably less to do with contact injuries resulting from high-speed collisions common in football. Therefore, the purpose of this study was to determine, over three consecutive collegiate football seasons, if a preseason total FMS score, the individual FMS test item scores, or the number of asymmetries can predict noncontact injury by position group. We expected to find an inverse relationship between FMS scores and injury, meaning that an increase in a FMS score corresponds with a decreased injury rate.

## METHODS

### Participants

Participants included all members of a Division I NCAA football team who were on the roster for one or more of the three consecutive years of data collection. This study included 208 male athletes. Mean age was 21.4 ± 1.84 yr (range of 18–26 yr). Participation per year: year 1, $n$ = 108 (height, 1.90 ± 0.06 m; weight, 105.19 ± 19.68 kg); year 2, $n$ = 119 (height, 1.88 ± 0.06 m; weight, 103.72 ± 19.53 kg); year, 3 $n$ = 116 (height, 1.87 ± 0.07 m; weight, 104.05 ± 19.62 kg). Because this study covered three consecutive years, some subjects were observed over multiple years, resulting in 343 total observations. The number of subjects observed for only 1 yr was 95; for 2 yr, 91; and for over the entire 3 yr of this study, 22. This study was approved by Brigham Young University's Institutional Review board (No. 17210) and was determined to be exempt from requiring written informed consent.

### Design

This was a 3-yr prospective descriptive epidemiology study, which evaluated the relationship between the total FMS score and injury rate in a NCAA football team. Secondary analyses utilized the scores found on the seven individual FMS tests and number of asymmetries and correlated them with injury.

### Procedures

Each participant had to pass a preparticipation physical and have any previous injuries reviewed by the sports medicine staff, who also performed a thorough musculoskeletal screen. Any individuals with orthopedic injuries over the prior year, or who demonstrated asymmetries in range of motion or strength were additionally screened by an orthopedic surgeon to be cleared for unrestricted participation and to be added to the roster. It should be noted that all offensive linemen wore bilateral knee braces during every exposure, and all subjects in each group were required to either tape their ankles or wear ankle braces for every practice or game. This is a fairly standard precaution utilized across all athletes in Division I football.

All participants were allocated into one of three groups based on their playing position. This was done to accommodate for the large differences in size, strength, speed, and the level of contact that exist in the various position groups during both practice and games. Groups with similar demands were combined, and all results are expressed in terms of these more homogeneous groups. These groups are labeled as "big," "combo," and "skill," The big group consisted of subjects playing on the offensive and defensive lines. The group labeled combo included those athletes playing linebacker and tight end. The remaining skill group included defensive backs, kickers, receivers, quarterbacks, and running backs.

The day after being cleared, all rostered players were assigned a time for FMS testing. All testing occurred on the same day, 1 d before the start of the preseason practice. Testing was done in groups using the standard 4-point scoring system with a range of 0–3 for the entire FMS screen using the protocol established by Cook (14,15,23). The FMS has been reported to demonstrate moderate to excellent intrarater (intraclass correlation coefficient = 0.74–0.92) and interrater (intraclass correlation coefficient = 0.76–0.98) reliability for testers ranging from novice to expert for each of its measured components and in both live and video-taped analysis (24–28). All measurements for hand length, footedness, and tibial tuberosity height were performed by the primary investigator before testing and were prepopulated on each subjects' data collection sheet. Each year, seven certified FMS testers performed one of the seven FMS screens on all subjects for that year. All testers had multiple years of experience performing these tests on athletic populations. This method has been found to be the most efficient way of testing large groups (29,30). Tests were all administered in the order that they are presented on the standard FMS scoring sheet (23).

We tracked participation in practice or games each day for all participants for the three consecutive collegiate football seasons. Injuries were recorded daily after each practice or game for the duration of this study. When data collection concluded, each of the three seasons' injuries was coded using the appropriate *International Classification of Diseases, Tenth Revision* code (31) for any injuries incurred. Athletic trainers who covered the practice or game documented mechanisms for all injuries to differentiate between contact and noncontact





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injuries. Data were deidentified and added to an encrypted database. We used season long data in aggregate form to report team and group averages to examine the relationship between FMS scores and injuries. The total score and each individual test item score were used to see if correlations existed with the noncontact injuries sustained over the course of each competitive season. Total injuries were reported and then assessed based on whether the injury was noncontact in nature. The number of IE was adjusted to reflect any decreases in participation during practices or games due to circumstances, such as contact injuries, disciplinary measures, athletes transferring, not playing in games, or sickness.

## Operational Definitions

For the current study, we used the following operational definitions.

### Injury

A musculoskeletal noncontact injury was defined as an injury that met each of the following criteria:

1. The injury occurred as a result of participation in an organized on or off field practice or competition.
2. The injury required consultation with a member of the medical team (e.g., certified athletic trainer, physical therapist, or physician).
3. The injury resulted in the inability to fully participate in a subsequent practice or competition.

Injuries are classified as contact or noncontact based on their mechanism.

### Injury Types

Contact: an injury that occurs with the mechanism being primarily a collision with another player or object

Noncontact: any injury that occurs where the mechanism or primary force causing the injury is of an intrinsic nature

### Injury Exposure (IE)

One IE represents one athlete participating in either one practice or game regardless of the time (number of plays or exercise repetition) associated with that participation (1,32,33). Not all study participants had the same number of IE, even if they were healthy throughout the entire season. For example, some players were not on the travel list and thus did not travel or participate in away games. Other athletes who did travel did not all play in the same number of plays; so even though all subjects had the same exposure to injury during the week of practice, their actual number of plays and exposure could vary significantly based on if and how much they played during games. Game injury rate is reported to be over nine times higher than regular season practice injury rate (34). Therefore, we did not count games as an exposure unless the subject participated in the game. This methodology matches that used in many epidemiological studies to differentiate between practice and game injuries (1,33,34).

### IE Rate

IE rate is the percentage of possible practices and games participated in divided by the total possible number of practices and games.

### Positional Groupings of Players

Big: offensive and defensive linemen

Skill: receivers, quarterbacks, running backs, defensive backs, and kickers

Combo: linebackers and tight ends

### Statistical Analysis

We used a generalized linear mixed-model methodology as implemented in the glmer() function from the lme4 package of R to analyze the data. For the percentage of plays and number of injuries per exposure response variables, we used the binomial family with the logit link function. For the raw number of injuries response variable, we used the Poisson family with a log link function. Because multiple players played more than one season, analyses needed to account for both within subject and between-subject error. We used generalized linear mixed models assuming either a binomial family (with a logit link function) or a Poisson family (with a log link function). With these models, the appropriate test statistics are normally distributed, so no adjustment for sample size was necessary. The analysis results also allowed us to determine positive or negative relationships based on the slope of the coefficient of

TABLE 1. Demographics by position group and whole team.

| | Year | Big | Combo | Skill | Team | Team FMS | Team Plays Game |
|---|---|---|---|---|---|---|---|
| Year | 1 | 35 | 24 | 49 | 108 | 16.0 ± 1.7 | 2059 |
| | 2 | 35 | 29 | 55 | 119 | 16.7 ± 1.7 | 1792 |
| | 3 | 32 | 22 | 62 | 116 | 14.8 ± 1.9 | 1770 |
| | Total | 102 | 75 | 166 | 343 | | 5621 |
| Height, m | | 1.92 ± 0.05 | 1.90 ± 0.05 | 1.85 ± 0.06 | 1.88 ± 0.07 | | |
| Mass, kg | | 130.31 ± 13.53 | 104.72 ± 7.11 | 90.02 ± 7.87 | 103.98 ± 19.55 | | |
| Injuries per year | 1 | 17 | 11 | 38 | 66 | | |
| | 2 | 9 | 10 | 24 | 43 | | |
| | 3 | 4 | 5 | 17 | 26 | | |
| | Total | 30 | 26 | 79 | 135 | | |
| No. asymmetries | 1 | 36 | 30 | 43 | 109 | | |
| | 2 | 38 | 30 | 50 | 118 | | |
| | 3 | 37 | 34 | 77 | 148 | | |
| | Total | 111 | 94 | 170 | 375 | | |



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the covariates. The α was set at 0.0167 to account for multiple comparisons.

## RESULTS

Participant demographics, total number of players by year and position, total number of team plays by year, number of noncontact injuries observed each year by group, number of asymmetries, and total mean team FMS scores by year are presented in Table 1.

Figure 1 shows the distribution of total FMS scores by year for each of the groups. Figure 2 demonstrates the yearly change in overall FMS scores for the 22 players who played for three consecutive years.

Table 2 contains the overall injury types and frequency by year across all 3 yr. Repeated injuries within the same athlete occurred as follows: year 1, six athletes sprained the same ankle twice, two athletes sprained the same ankle three times, and one athlete pulled a hamstring twice; year 2, one athlete pulled a hamstring twice and one athlete had a disk herniation that resulted in three separate episodes of missed participation; and year 3, one athlete suffered from a hip labral tear, causing him to miss participation two separate times, one athlete strained his hamstrings twice, and another athlete strained his hamstrings three times in the course of the year, causing missed participation. In between injuries, all athletes were fully cleared by the team physician and participated in practice and games.

Overall, we found that total FMS score for the team as a whole was not a significant indicator of the number of injuries experienced over the 3 yr studied ($P = 0.130$). When considered for the different groups, the total FMS score was also not a significant indicator of the number of injuries for the big ($P = 0.196$), combo ($P = 0.172$), or skill ($P = 0.190$) groups. Significance did not change when IE were taken into consideration: whole team ($P = 0.128$), big ($P = 0.253$), combo ($P = 0.237$), or skill ($P = 0.187$).

Overall, we found that the total amount of asymmetries for the entire team was not a significant indicator of the number of injuries that were experienced over the 3 yr studied ($P = 0.472$). When considered for the different groups, the total number of asymmetries was also not a significant indicator of the number of injuries for the big ($P = 0.148$), combo ($P = 0.475$), or skill ($P = 0.792$) groups. Significance does not change when IE are taken into consideration: entire team ($P = 0.508$), big ($P = 0.129$), combo ($P = 0.464$), or skill ($P = 0.707$). Table 3 shows the individual test FMS scores by years and groups. Table 4 contains the relationship between individual FMS test item scores and the number of injuries observed. No individual test item was significant in the big group. In the combo group, there was a significant inverse relationship for push-up stability and injury prevalence. The rotary stability test item score was found to be significant with respect to the total number of injuries in the skill and for the total team. This relationship was positive, meaning an improved score in the rotary stability test is associated with an increased frequency of injury. No substantial changes in the relationship of FMS test item and number of injuries were observed when corrected for IE by group and total team.

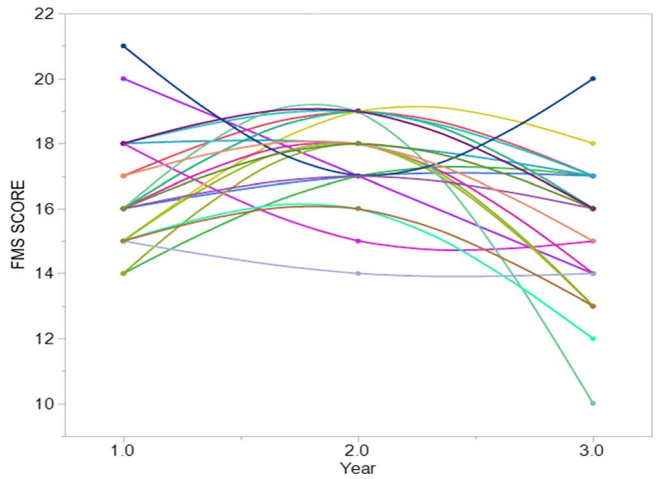


**FIGURE 2—Total FMS scores for players tested all 3 yr.**

## DISCUSSION

This study is the first multiyear prospective study designed to assess the FMS's ability to predict noncontact injury in Division I NCAA football players.

**The relationship of total FMS scores to injury.** Our data demonstrated a lack of any significant correlation between Total FMS score and total number of injuries even when controlled for IE. In addition, no correlations were noted for the big, combo, or skill groups, even when controlled for IE.

These results are in line with other studies that have reported no relationship between scores and injury prediction (19,20).

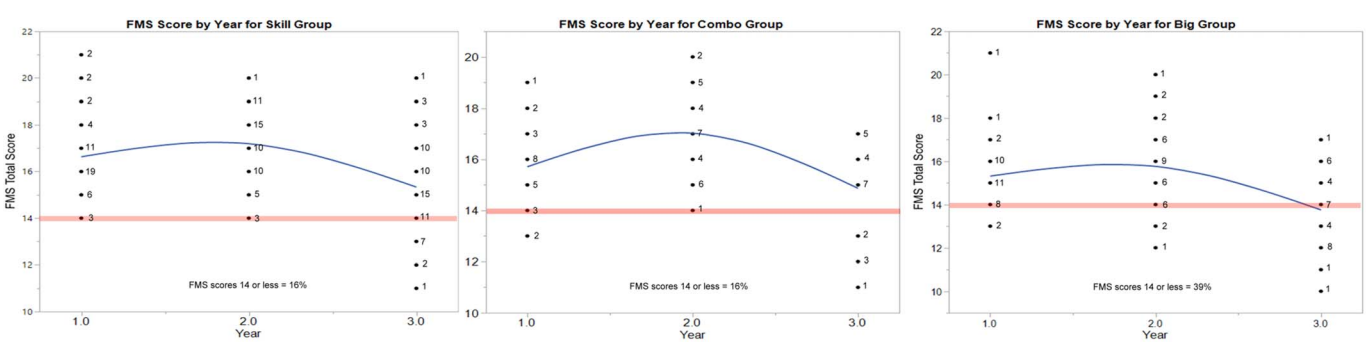


**FIGURE 1—Total FMS scores by year and by group.**



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For example, Mokha et al. (19) found that Division II athletes (rowers, volleyball players, and soccer players) with an FMS score of 14 and lower did not have a greater injury risk compared with those with a higher score. Yeung et al. (20) assessed injury rate of the lower extremity in an English Football Championship team over a whole season. However, their study had only 16 participants, and the results might not be representative of American football players (Division I or professional). None of these studies made a distinction between contact and noncontact injuries. We believe that making this distinction is an important consideration because noncontact injuries are likely more a function of movement quality (or lack thereof) than injuries resulting from high-speed collisions or other contact with another football player.

On the other hand, studies have claimed that functional movement patterns, as identified by the FMS, are predictive as identifiable risk factors for injury (16,17). The first study (16) ($n$ = 46) established a cutoff point in the FMS score; that is, a score of 14 or less was positive to predict serious injury. It is worth noting that the authors defined "serious injury" as an injury that placed the injured player "on the injured reserve" and that necessitates a playing time loss of 3 wk. The second study (17) ($n$ = 238) was set out to validate this cutoff point but was not able to accomplish this goal. Instead, it showed that having at least one asymmetry on the FMS, regardless of a player's composite score, increased the injury risk. It is possible that their inability to validate the initial study was due to the change in their definition of injury. The second study defined injury as "any time loss from either practice or preseason game"; a definition we adopted for our present study was the exception that we collected data over three complete seasons. The first study (16) collected data over one season, whereas the other study (17) collected data only during preseasons. Both of the aforementioned studies were performed using professional American football players, representing less than the top 2% of NCAA players. Thus, the FMS may be a more useful tool to predict injury in elite players. An additional important difference between those and our study was that there was no distinction made between contact and noncontact injury.

TABLE 2. Injury type and frequency by year.

| 1st Year Injury | Count | 2nd Year Injury | Count | 3rd Year Injury | Count |
|---|---|---|---|---|---|
| Abdominal strain | 1 | Achilles strain | 2 | ACL rupture | 3 |
| ACL rupture | 1 | ACL rupture | 3 | Ankle sprain | 1 |
| Groin adductor strain | 2 | Ankle sprain | 7 | Elbow hyperextension | 1 |
| Ankle sprain | 29 | Gastrocnemius strain | 1 | Hamstring strain | 12 |
| Articular cartilage knee | 1 | Groin adductor strain | 2 | Hip flexor strain | 1 |
| Avulsion fracture ASIS | 1 | Hamstring strain | 9 | Hip labral tear | 2 |
| Chondrolysis hip | 1 | High ankle sprain | 1 | Lumbar disk herniation | 1 |
| Forefoot sprain | 1 | Hip flexor strain | 1 | Lumbar strain | 1 |
| Hamstring strain | 11 | Knee edema | 1 | MCL strain | 1 |
| Hip flexor strain | 3 | Lisfranc strain | 1 | Pars stress fracture | 1 |
| Lumbar disk herniation | 3 | Lumbar disk herniation | 5 | Quadriceps strain | 1 |
| Lumbar facet | 1 | Lumbar strain | 1 | Triceps strain | 1 |
| Lumbar strain | 3 | MCL sprain | 3 | | |
| Meniscus | 2 | Metatarsal contusion | 1 | | |
| Metatarsal Fx | 2 | Patellar tendonitis | 2 | | |
| Patellar tendonitis | 1 | Pectoral tear | 1 | | |
| Plantar fascia strain | 1 | | | | |
| Synovitis knee | 1 | | | | |
| Turf toe | 1 | | | | |
| Total | 66 | | 43 | | 26 |

Fx, fracture; MCL, medial collateral ligament.

TABLE 3. Test item FMS scores by group and total team.

| | Big | Combo | Skill |
|---|---|---|---|
| Year 1 | | | |
| Players | 35 | 24 | 49 |
| Deep squat | 2.14 ± 0.43 | 2.13 ± 0.34 | 2.18 ± 0.44 |
| Hurdle step | 1.94 ± 0.42 | 2.04 ± 0.36 | 2.22 ± 0.47 |
| In-line lunge | 2.11 ± 0.32 | 2.13 ± 0.34 | 2.25 ± 0.43 |
| Shoulder mobility | 2.00 ± 0.69 | 2.08 ± 0.88 | 2.22 ± 0.74 |
| Active straight leg raise | 2.51 ± 0.56 | 2.42 ± 0.50 | 2.78 ± 0.42 |
| Push-up stability | 2.66 ± 0.64 | 2.88 ± 0.34 | 2.90 ± 0.31 |
| Rotary stability | 1.91 ± 0.37 | 2.04 ± 0.36 | 2.08 ± 0.28 |
| FMS™ total | 15.31 ± 1.49 | 15.71 ± 1.52 | 16.63 ± 1.62 |
| Year 2 | | | |
| Players | 35 | 29 | 55 |
| Deep squat | 2.14 ± 0.43 | 2.59 ± 0.50 | 2.55 ± 0.50 |
| Hurdle step | 2.31 ± 0.47 | 2.17.38 | 2.26 ± 0.44 |
| In-line lunge | 2.63 ± 0.49 | 2.55 ± 0.51 | 2.73 ± 0.45 |
| Shoulder mobility | 2.11 ± 0.72 | 2.35 ± 0.72 | 2.27 ± 0.71 |
| Active straight leg raise | 2.31 ± 0.47 | 2.48 ± 0.51 | 2.62 ± 0.53 |
| Push-up stability | 2.51 ± 0.70 | 2.86 ± 0.35 | 2.82 ± 0.39 |
| Rotary stability | 1.74 ± 0.44 | 2.03 ± 0.19 | 1.96 ± 0.19 |
| FMS™ total | 15.77 ± 1.80 | 17.03 ± 1.68 | 17.20 ± 1.51 |
| Year 3 | | | |
| Players | 32 | 22 | 63 |
| Deep squat | 1.81 ± 0.54 | 1.96 ± 0.38 | 2.05 ± 0.49 |
| Hurdle step | 2.03 ± 0.31 | 2.00 ± 0.31 | 2.19 ± 0.40 |
| In-line lunge | 1.91 ± 0.39 | 2.27 ± 0.46 | 2.18 ± 0.38 |
| Shoulder mobility | 2.03 ± 0.82 | 2.18 ± 0.85 | 2.19 ± 0.72 |
| Active straight leg raise | 2.00 ± 0.842 | 2.09 ± 0.81 | 2.41 ± 0.71 |
| Push-up stability | 2.81 ± 0.40 | 2.91 ± 0.29 | 2.89 ± 0.32 |
| Rotary stability | 1.16 ± 0.37 | 1.46 ± 0.51 | 1.429 ± 0.50 |
| FMS™ total | 13.75 ± 1.76 | 14.86 ± 1.89 | 15.33 ± 1.87 |
| Total team | | | |
| Players | 102 | 75 | 167 |
| Deep squat | 2.04 ± 0.49 | 2.25 ± 0.50 | 2.25 ± 0.52 |
| Hurdle step | 2.10 ± 0.43 | 2.08 ± 0.36 | 2.22 ± 0.43 |
| In-line lunge | 2.223 ± 0.51 | 2.33 ± 0.48 | 2.38 ± 0.49 |
| Shoulder mobility | 2.05 ± 0.74 | 2.21 ± 0.81 | 2.23 ± 0.72 |
| Active straight leg raise | 2.28 ± 0.67 | 2.35 ± 0.63 | 2.59 ± 0.59 |
| Push-up stability | 2.66 ± 0.61 | 2.88 ± 0.33 | 2.87 ± 0.34 |
| Rotary stability | 1.62.51 | 1.87 ± 0.45 | 1.80 ± 0.46 |
| FMS™ total | 14.98 ± 1.88 | 15.97 ± 1.90 | 16.33 ± 1.86 |

**The relationship of individual test item FMS scores to injury.** The only individual test item that was predictive of injury for the entire team was the rotary stability ($P < 0.001$) and also for the skill group ($P$ = 0.011). However, the

TABLE 4. Relationship between test item FMS scores and number of injuries by group and total team.

| | Group | | | |
|---|---|---|---|---|
| | Big | Combo | Skill | Total Team |
| Total Scores | *P* | *P* | *P* | *P* |
| Deep squat | 0.087(+) | 0.515(–) | 0.197(+) | 0.054(+) |
| Hurdle step | 0.739(+) | 0.904(–) | 0.071(+) | 0.062(+) |
| In-line lunge | 0.179(+) | 0.148(–) | 0.543(+) | 0.396(+) |
| Shoulder mobility | 0.210(–) | 0.224(–) | 0.550(–) | 0.123(–) |
| Active straight leg raise | 0.358(+) | 0.996(–) | 0.862(–) | 0.537(+) |
| Push-up stability | 0.933(–) | 0.001(–)* | 0.943(+) | 0.678(–) |
| Rotary stability | 0.054(+) | 0.297(+) | 0.011(+)* | <0.001(+)* |

* denotes significance; (+) or (–) denotes direction of the relationship.



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correlation between the two variables was positive, suggesting that, as the rotary stability test score increases, the rate of injury increases as well. This relationship was the same when controlled for IE. Such a relationship is contrary to the underlying premise of the FMS. The skill group, which consists of receivers, quarterbacks, running backs, defensive backs, and kickers, is more frequently exposed to injury-producing noncontact situations as their position requires repeated running, cutting and evasive maneuvers, and high-speed extremity movements that exceed the rigid confines of the stationary rotary stability component of the FMS. The rotary stability test is performed in a relative static quadruped position and thus may not translate into a predictive measure for dynamic movement. In addition, it should be noted that the number of asymmetries among all the tests was double in the skill group. Perhaps this is a reflection of the unilateral skills (e.g., kicking and throwing) and movement patterns that are necessary or advantageous for athletes in this group.

In the combo group, the only test that was predictive of injury was push-up stability for both the total number of injuries and injuries weighted for IE. Risk analysis was performed on these two significant tests to determine the expected change in risk associated with a 1-unit change in the push-up stability test score. In the case of total number of injuries as a function of push-up stability score, the two possible scores that corresponded with the risk of injury was a score of 2 or 3. When the score was 2, the risk of injury was 13.44%; when the score improved to a 3, risk was reduced to 3.67%. When we looked at the same test in relation to the number of injuries, controlled for IE, we found a risk of 11% associated with a score of 2 and a reduction in risk to 3.2% with a score of 3. Thus, it may be worth improving shoulder stability to reduce the risk of injury in this subset of players.

**The relationship between number of FMS asymmetries and injury.** Several studies have suggested that the number of asymmetries demonstrated in the five individual tests that measure both sides of the body is an additional predictor of injury (17,19). Our analysis revealed no statistically significant relationship between the total number of injuries and the number of asymmetries for the total team or by group. This relationship did not change when we controlled for IE. It is worth noting that the majority of our findings were negative relationships, suggesting that more asymmetries lead to a decreased risk of injury. Athletes often exhibit discipline-specific asymmetries that might be necessary to excel in their particular sport (35).

**Between-year changes in FMS score by groups.** The FMS scores by group demonstrated an interesting transition over the duration of the study. In the first 2 yr of our study, the training was primarily focused on flexibility, explosiveness, and body weight exercises. This approach was reflected in improved total average FMS scores in all groups (year 1: 16.63 ± 1.62 for skill, 15.71 ± 1.52 for combo, 15.31 ± 1.49 for big; year 2: 17.20 ± 1.51 for skill, 17.03 ± 1.68 for combo, 15.77 ± 1.80 for big). In the third year, a new strength and conditioning staff was hired. They placed a much greater emphasis on sheer strength and heavy lifting. This change in training philosophy may be the reason for the lower average FMS scores in year 3 (skill, 15.33 ± 1.87; combo, 14.86 ± 1.89; big, 13.75 ± 1.76). The changes in total FMS scores from years 1 to 2 and from 2 to 3 seem to support the findings of Kiesel et al. (36), who reported that FMS scores changed with specific training.

**Limitations.** One potential limitation is that the coaching staff changed after 2 yr into our 3-yr data collection period. Furthermore, the strength and conditioning staff changed each of the 3 yr studied. This resulted in a change in the offseason training and in-season practice characteristics from year to year. Specifically, for the first 2 yr of this study, the team utilized a "go hard, go fast" mentality in both games and practice settings (total of 2059 plays). The third year was based on a more traditional, slower paced game, where the players saw significantly fewer overall plays (total of 1770 plays) in both the practice and game settings. Likely, this had an effect on total FMS scores. The graphs in Figure 2 seem to confirm this supposition as they show little consistency over time. This could be construed as a limitation; however, in the normal year-to-year flow of collegiate football, it is common for coaching staffs and philosophies to change. For example, there was an average of just over 23 head coaching changes per year over the past 7 yr among the Division I Football Bowl Series teams (37). Therefore, although a coaching change might be considered a limitation, the fact that it happened during the course of this study actually gives a more realistic representation of the changes collegiate athletes face over the course of their collegiate football careers.

Because statistical significance is related to sample size, and the sample size for our data set was quite large, it was likely that we would get results that were statistically significant but that had no practical significance. That is, we might show a statistically significant relationship that was meaningless for use in a practical or clinical application.

## CONCLUSIONS

Although the FMS can be used to assess the quality of fundamental movement and demonstrates asymmetries for five of the seven item tests, overall our data suggest that it is not a valid tool for predicting noncontact injury in NCAA Division I football. We established normative data for multiple seasons of intercollegiate football for total FMS and individual test item scores. Although we found several measures that seemed to display statistical significance, the relationship between statistical significance and practical significance must be considered. Based on our findings, we suggest that other assessment tests be considered or developed that better predict injury of intercollegiate football players.

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The results of this study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine.

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